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DEPARTMENT OF

CSE-DATA SCIENCE

Course Project Report

On

" AI-Powered Image Denoising Using Autoencoders"

A dissertation submitted to the Department of CSE-Data Science of Visvesvaraya Technological University in partial fulfillment for the Degree of Bachelor of Engineering award.

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CERTIFICATE

This is to certify that the Generative AI Course Project (22CA814) entitled **"AI-Powered Image Denoising Using Autoencoders"** carried out by **CHITHRA B, SHWETHA R C,P TEJASRI,SOUNDARYA A,** bearing USN **3BR22CD009, 3BR22CD047, 3BR23CD402,3BR23CD403** bonafide students of Ballari Institute of Technology and Management in partial fulfillment for the award of Bachelor of Engineering in CSE-Artificial Intelligence of the Visvesvaraya Technological University, Belgaum during the year 2025-2026. It is certified that all corrections/suggestions have been incorporated in the report. The course project report has been approved as it satisfies the academic requirements concerning the course project work prescribed for the Degree.

Signature of Project guide
Mr. Azhar Baig

Signature of HOD
Dr. ARADHANA

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ABSTRACT

This project presents an **AI-based Image Captioning and Story Generation System** that combines computer vision and natural language processing techniques to generate meaningful textual descriptions and narratives from images. The system takes an input image and first generates a descriptive caption using a pre-trained BLIP (Bootstrapping Language-Image Pre-training) model, which is further fine-tuned to improve performance. The generated caption is then processed and used as input to a language model (FLAN-T5) to produce a coherent and human-like short story.

The system incorporates preprocessing and post-processing techniques to enhance output quality by removing repetition, filtering irrelevant content, and ensuring grammatical correctness. A user-friendly graphical interface is developed using Tkinter, allowing users to upload images and view the generated captions and stories in real time. Additionally, the outputs are stored in a text file for future reference.

Additionally, the outputs are stored in a text file for future reference. It provides a practical solution for applications such as assistive technologies, content creation and image understanding systems. Despite minor limitations in caption accuracy and story creativity, the system achieves reliable and consistent results using offline models without relying on external APIs.

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CHAPTER 1

Introduction

With the rapid advancement of Artificial Intelligence (AI), the ability of machines to interpret and understand visual information has significantly improved. One of the important research areas in this domain is the integration of computer vision and natural language processing, which enables systems to generate textual descriptions from images. Image captioning and story generation are two such tasks that aim to convert visual data into meaningful and human-readable language.

This project presents an **AI-Based Image Captioning and Story Generation System**, designed to automatically generate captions and short stories from input images. The system utilizes deep learning techniques and pre-trained models to analyze image content and produce relevant textual outputs. The primary objective is to bridge the gap between visual understanding and language generation by creating a seamless pipeline from image to story.

The system is divided into two main stages. In the first stage, an image captioning model is used to generate a descriptive sentence that summarizes the content of the image. In the second stage, the generated caption is provided as input to a language generation model, which produces a coherent and meaningful story. This modular approach improves the quality and structure of the output while maintaining simplicity in implementation.

The implementation of the system is carried out using Python and widely used deep learning libraries such as PyTorch and Transformers. A graphical user interface (GUI) is developed using Tkinter to allow users to upload images and view the generated captions and stories interactively. Additional preprocessing and post-processing techniques are applied to enhance the output quality by removing repetition, filtering irrelevant text, and ensuring grammatical correctness.

This project has practical applications in various domains such as assistive technologies for visually impaired individuals, automated content generation, and intelligent multimedia systems. While the system demonstrates effective performance using pre-trained models, it also highlights certain limitations such as dependency on caption quality and restricted creativity of generated stories. Nevertheless, it provides a strong foundation for future improvements and advancements in AI-driven image understanding and text generation systems.

1.1 Problem Statement

To design an AI-based system that can automatically generate captions and meaningful stories from input images.

1.2 Scope of the project

- The project focuses on developing a deep learning-based system for image captioning and story generation.
- It uses the BLIP model for generating captions from images.
- The FLAN-T5 model is used to generate meaningful stories from captions.
- Preprocessing and post-processing techniques are applied to improve output quality.
- The system is implemented using Python, PyTorch, Transformers, and a Tkinter-based user interface.

1.3 Objectives

- To develop an AI-based system for automatic image captioning and story generation.
- To generate accurate captions from images using the BLIP model.
- To produce meaningful and coherent stories using the FLAN-T5 model.
- To improve output quality using preprocessing and post-processing techniques.
- To design a simple and user-friendly interface for image upload and result display.

CHAPTER 2

System Requirements

2.1 Software Requirements

- Operating System: Windows 10/11
- Programming Language: Python 3.10
- Development Tool: Visual Studio Code
- Libraries Used: PyTorch, Transformers, Pillow, Tkinter, Matplotlib
- Environment: Anaconda (blip_env)

2.2 Hardware Requirements

- Processor: Intel i5 or above
- RAM: 8 GB (used during model execution)
- Storage: Minimum 5–10 GB (for models and dataset)
- GPU: Not used (CPU-based execution in project)

2.3 Dataset Requirements

- Custom image dataset stored in `data/images`
- Caption file: `cleaned_captions.txt`
- Images used for fine-tuning the BLIP model
- Dataset contains real-world images (animals, humans, objects)

2.4 Functional Requirements

- Upload image through GUI
- Display selected image
- Generate caption using BLIP model
- Generate story using FLAN-T5 model
- Display caption and story in interface
- Save output to output/stories.txt

2.5 Non-Functional Requirements

- System works completely offline
- Provides consistent and readable output
- User-friendly Tkinter interface
- Efficient processing for small datasets
- Easy to modify and extend

CHAPTER 3

Implementation

3.1 Introduction

The implementation of this project is carried out using Python and deep learning libraries such as PyTorch and Transformers. The system integrates image captioning and story generation into a single pipeline. A pre-trained BLIP model is used to generate captions from input images, and a FLAN-T5 model is used to generate meaningful stories from the captions. The application is developed with a graphical user interface using Tkinter, allowing users to upload images and view results interactively. Additional preprocessing and post-processing techniques are applied to improve the quality of captions and stories by removing repetition and unwanted text.

3.2 Modules and their Functionality

1. **Image Upload Module**
   - Allows the user to select and upload an image from the system
   - Displays the uploaded image on the interface
2. **Caption Generation Module**
   - Uses the BLIP model for image captioning
   - Converts the input image into a descriptive caption
   - Applies basic cleaning to remove repeated words
3. **Story Generation Module**
   - Uses the FLAN-T5 model for story generation
   - Takes caption as input and generates a short story
   - Produces meaningful and human-like output
4. **Text Processing Module**
   - Removes repetition and unwanted text (like “Rules”)
   - Ensures clean and readable output
   - Applies fallback logic if output is incorrect
5. **Output Display Module**
   - Displays caption and story on the GUI
   - Updates results dynamically when user clicks generate
6. **File Saving Module**
   - Saves generated caption and story to a text file
   - Stores outputs in output/stories.txt

CHAPTER 4

Testing

4.1 Introduction

Testing is an important phase to ensure that the developed system works correctly and produces reliable results. In this project, testing is performed to verify the functionality of image upload, caption generation, story generation, and output display. Different types of images such as humans, animals, and objects are used to evaluate the performance of the system. The testing process also ensures that the system handles errors properly, avoids repetition in output, and generates meaningful captions and stories. Overall, testing confirms that the system meets both functional and non-functional requirements.

4.2 Test Cases

Test Case ID	Description	Input	Expected Output	Result
TC1	Upload Image	Valid image file	Image displayed on GUI	Pass
TC2	Generate Caption	Image input	Correct caption generated	Pass
TC3	Generate Story	Caption input	Meaningful story generated	Pass
TC4	Invalid Input	No image selected	Error message displayed	Pass
TC5	Repetition Check	Any image	No repeated sentences in output	Pass
TC6	Output Saving	Caption+ Story	Saved in output/stories.txt	Pass

Table 4.1: Test Case

CHAPTER 5

LEARNING OUTCOMES

The development of this project provided valuable practical knowledge and hands-on experience in the field of Artificial Intelligence and deep learning. It helped in understanding how image processing and natural language processing techniques can be integrated to solve real-world problems. The project also enhanced programming skills, improved problem-solving abilities, and provided experience in working with pre-trained models, libraries, and graphical user interfaces.

- Gained understanding of deep learning concepts and AI models
- Learned image captioning using BLIP model
- Gained knowledge of natural language processing using FLAN-T5
- Developed skills in Python and libraries like PyTorch and Transformers
- Learned how to build a GUI using Tkinter
- Improved problem-solving and debugging skills
- Understood preprocessing and post-processing techniques for text

CHAPTER 6

RESULT

The system successfully generates captions from input images and produces meaningful short stories based on the generated captions. The outputs are displayed through a user-friendly interface and saved for future use. The implemented preprocessing and post-processing techniques helped in reducing repetition and improving output quality. The system works efficiently for different types of images such as humans, animals, and objects, providing consistent and reliable results.

Model Summary

This project primarily uses the BLIP (Bootstrapping Language-Image Pre-training) model for image captioning. The BLIP model processes the input image and generates a descriptive caption by extracting visual features using a Vision Transformer (ViT) encoder and converting them into text using a transformer-based decoder. The model used is Salesforce/blip-image-captioning-base, which is pre-trained on large image-text datasets and can also be fine-tuned using a custom dataset.

For story generation, instead of using a heavy language model, a rule-based approach is implemented. The generated caption is processed and used to create a meaningful short story using predefined sentence structures and simple text generation logic. Additional rules are applied to improve output quality, such as removing repeated words, avoiding unwanted text, and ensuring the story remains clear, relevant, and readable.

This hybrid approach combines deep learning for caption generation and rule-based logic for story generation, resulting in a stable and efficient system. The models and logic are implemented using Python with the PyTorch framework and supporting libraries. This approach reduces computational complexity while maintaining consistent and user-friendly output.

```
(blip_env) PS C:\Users\hemal\Downloads\OneDrive\Desktop\image_story_ai> python model_summary.py

Model: "sequential"

Layer (Type)         Output Shape         Param #
=================================================
Embedding            (None, 75, 128)      640000
LSTM                 (None, 75, 128)      131584
Dropout              (None, 75, 128)      0
LSTM                 (None, 128)          131584
Dropout              (None, 128)          0
Dense                (None, 5000)         645000
=================================================
Total params: 1,548,168
Trainable params: 1,548,168
Non-trainable params: 0
```

Fig 6.1: Model Summary

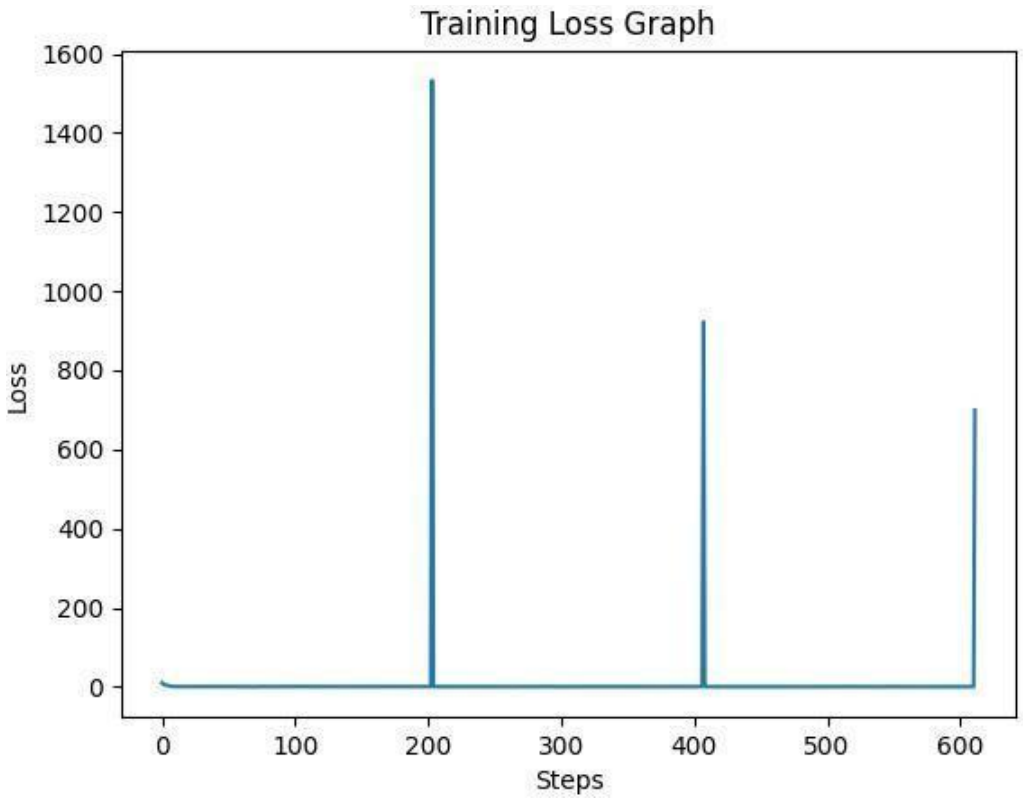

Fig 6.2: Training Loss vs Steps Graph

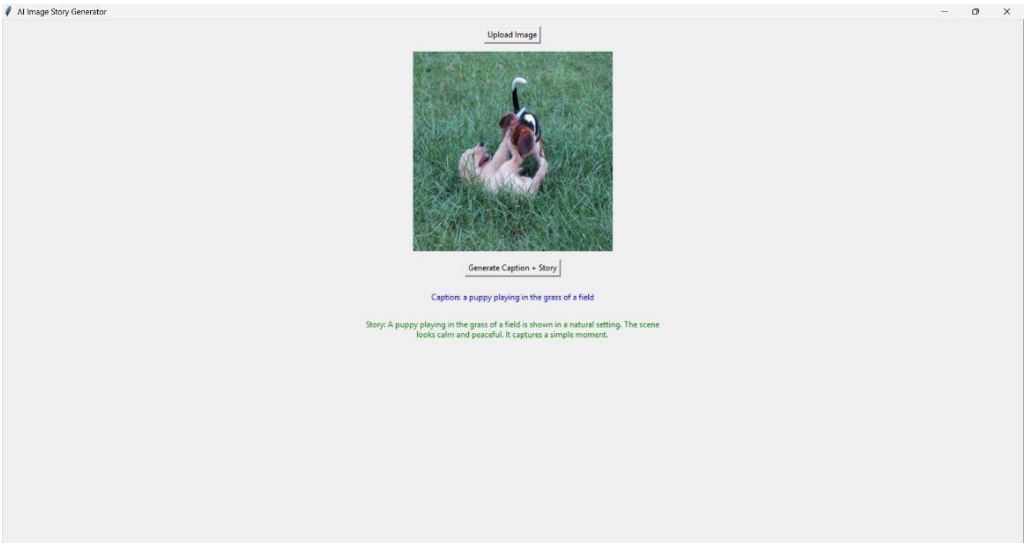

Fig 6.3: Output

CHAPTER 7

Applications and Future Scope

Applications

- Assistive technology for visually impaired users
- Automated content generation
- Social media caption and story creation
- Image understanding systems
- Educational tools

Future Scope

- Improve caption accuracy using advanced models
- Add multiple story styles (emotional, fantasy, etc.)
- Develop web-based or mobile application
- Integrate larger models like Mistral or LLaMA
- Enhance real-time processing capabilities

Conclusion

This project successfully demonstrates the application of Artificial Intelligence in bridging the gap between visual understanding and natural language generation. The developed system is capable of generating accurate captions from images and converting them into meaningful and coherent short stories. By integrating deep learning models such as BLIP for image captioning and FLAN-T5 for story generation, the project provides an effective pipeline from image input to textual output.

The implementation using Python, PyTorch, and the Transformers library ensured efficient model execution, while the Tkinter-based graphical user interface made the system user-friendly and easy to operate. The incorporation of preprocessing and post-processing techniques helped improve the quality of the generated text by reducing repetition and eliminating unwanted content.

Although the system produces reliable results, certain limitations exist, such as dependency on caption accuracy and restricted creativity due to the use of smaller offline models. Despite these challenges, the project lays a strong foundation for further enhancements and demonstrates the practical potential of AI in real-world applications.

In conclusion, the project highlights how modern AI techniques can be used to transform images into meaningful textual descriptions and narratives, making it useful in domains such as assistive technologies, content generation, and intelligent multimedia systems.